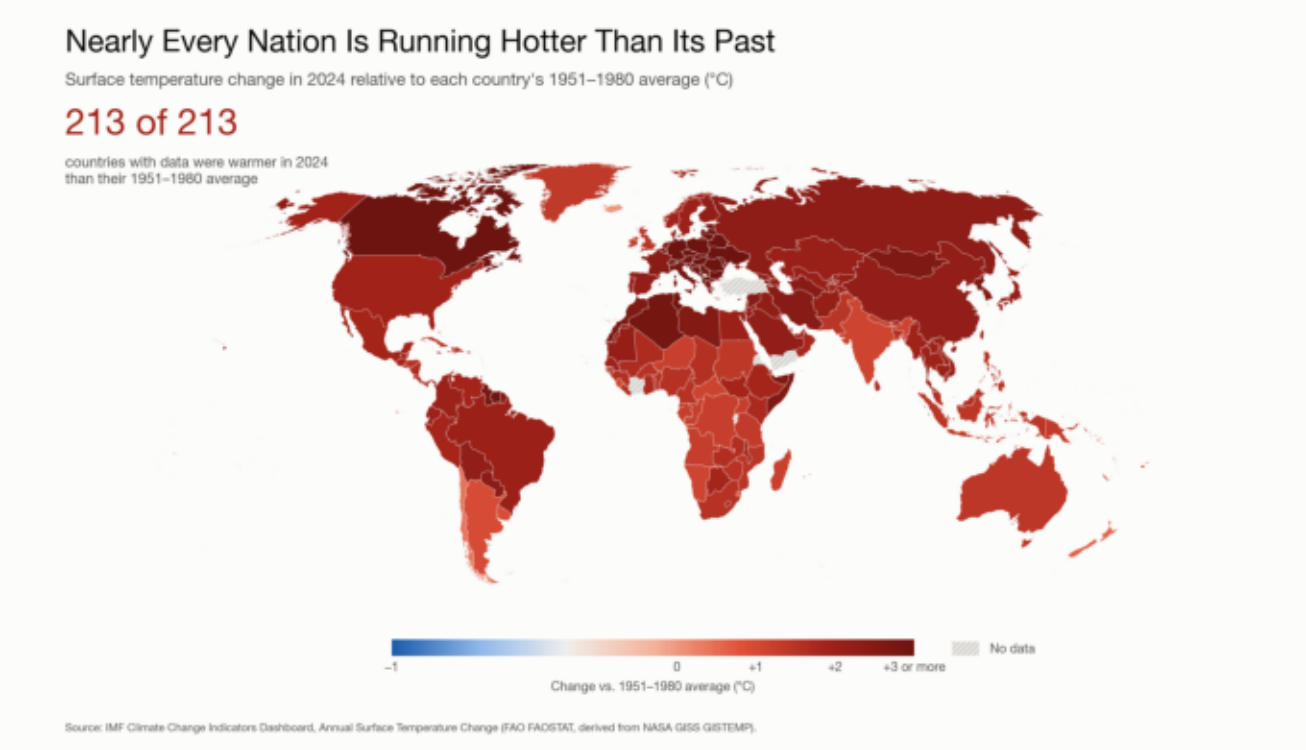
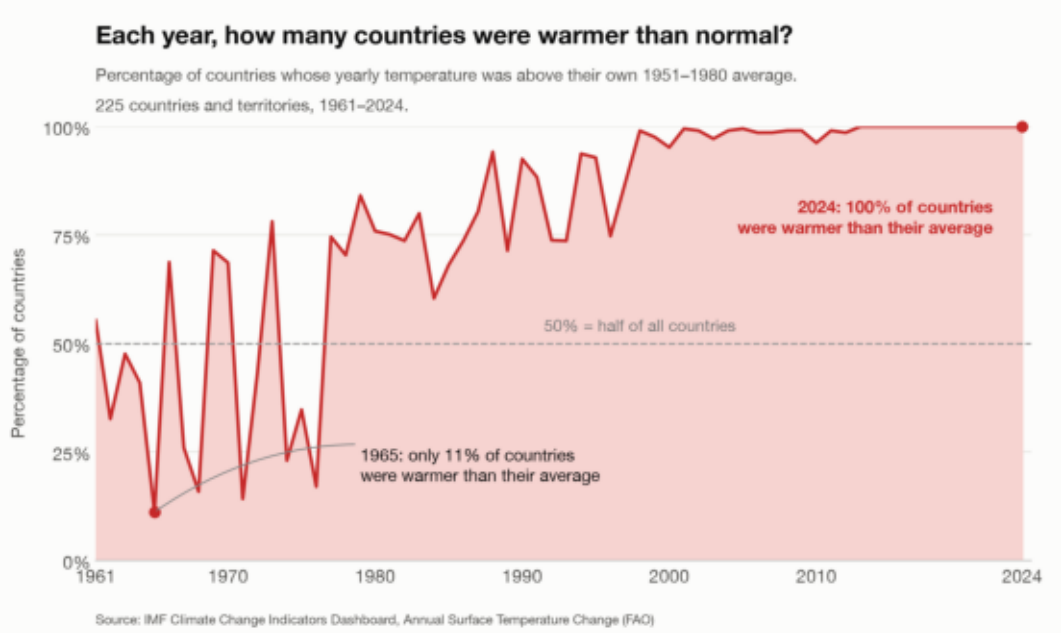
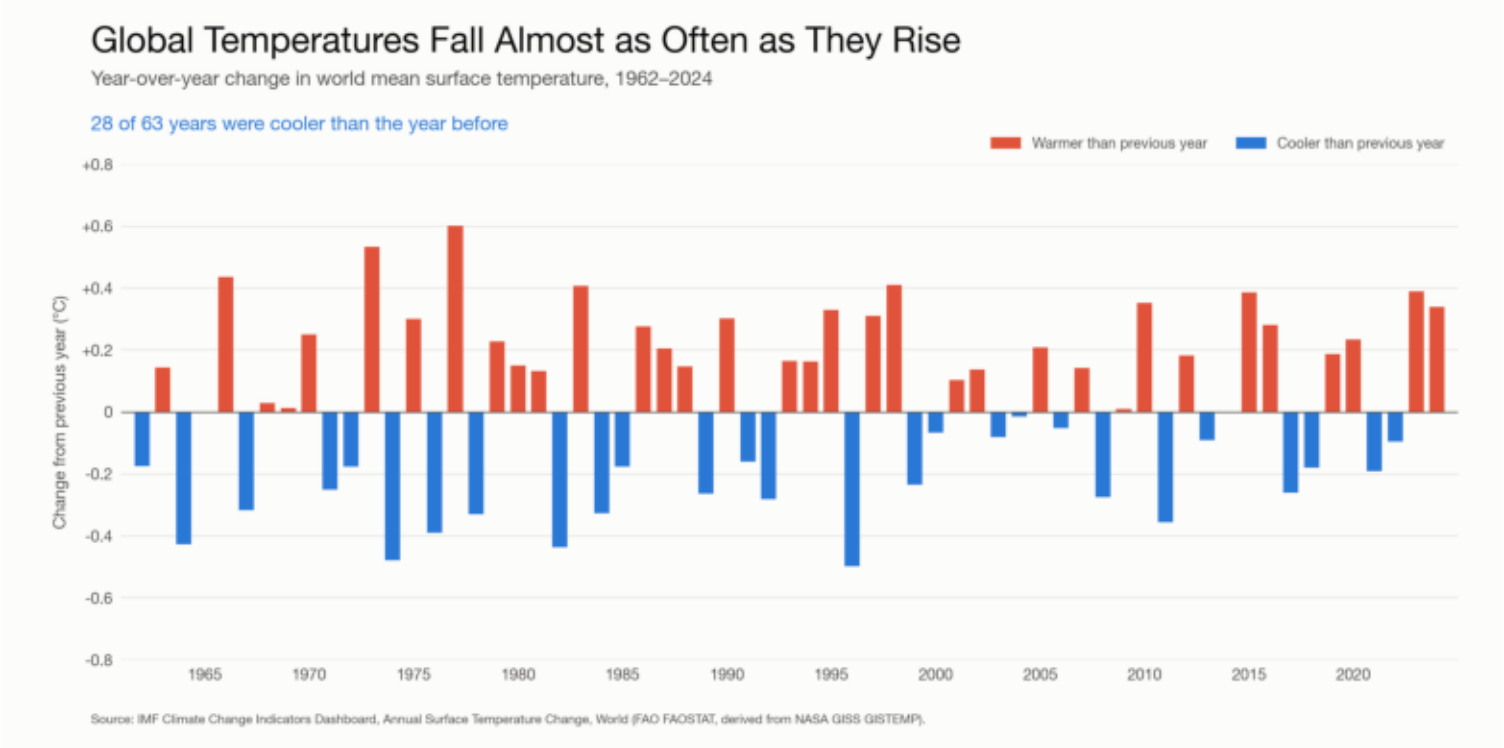
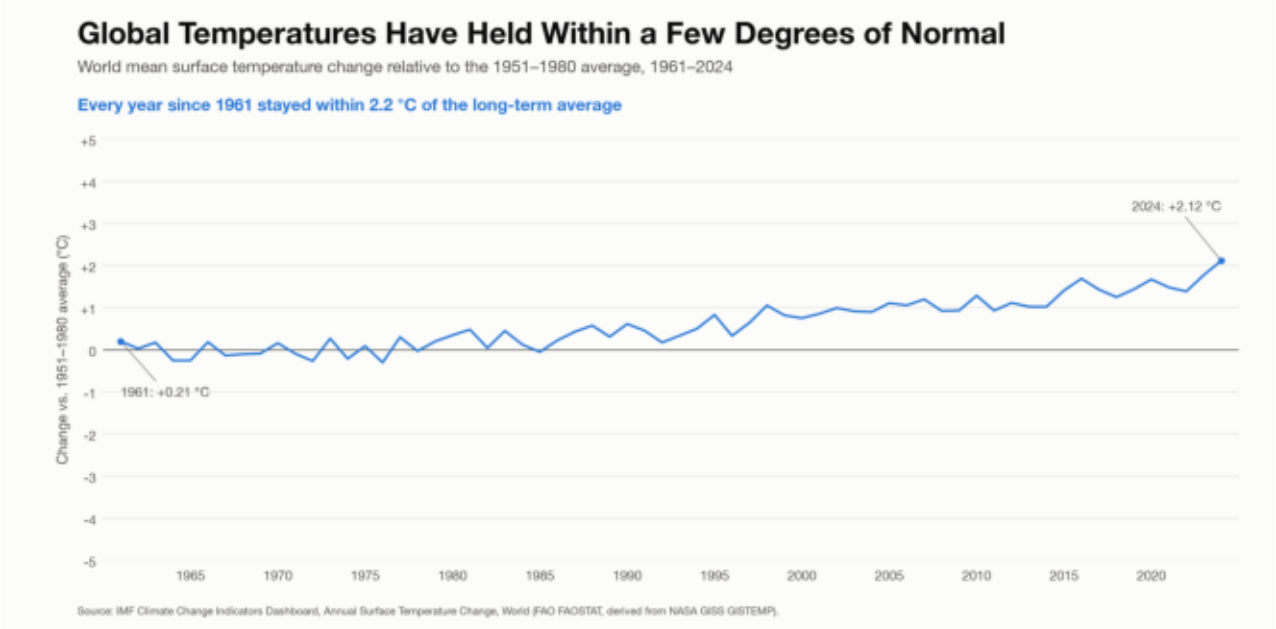
Set 1


Nearly Every Nation Is Running Hotter Than Its Past
Surface temperature change in 2024 relative to each country's 1951–1980 average (°C)
213 of 213
countries with data were warmer in 2024 than their 1951–1980 average
−1
0
+1
+2
+3 or more
No data
Change vs. 1951–1980 average (°C)
Source: IMF Climate Change Indicators Dashboard, Annual Surface Temperature Change (FAO FAOSTAT, derived from NASA GISS GISTEMP).


Each year, how many countries were warmer than normal?
Percentage of countries whose yearly temperature was above their own 1951–1980 average.
225 countries and territories, 1961–2024.
Percentage of countries
100%
75%
50%
25%
0%
1961
1970
1980
1990
2000
2010
2024
2024: 100% of countries were warmer than their average
50% = half of all countries
1965: only 11% of countries were warmer than their average
Source: IMF Climate Change Indicators Dashboard, Annual Surface Temperature Change (FAO)

Set 2


Global Temperatures Fall Almost as Often as They Rise
Year-over-year change in world mean surface temperature, 1962–2024
28 of 63 years were cooler than the year before
Warmer than previous year
Cooler than previous year
Change from previous year (°C)
+0.8
+0.6
+0.4
+0.2
0
-0.2
-0.4
-0.6
-0.8
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
Source: IMF Climate Change Indicators Dashboard, Annual Surface Temperature Change, World (FAO FAOSTAT, derived from NASA GISS GISTEMP).


Global Temperatures Have Held Within a Few Degrees of Normal
World mean surface temperature change relative to the 1951–1980 average, 1961–2024
Every year since 1961 stayed within 2.2 °C of the long-term average
Change vs. 1951–1980 average (°C)
+5
+4
+3
+2
+1
0
-1
-2
-3
-4
-5
2024: +2.12 °C
1961: +0.21 °C
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
Source: IMF Climate Change Indicators Dashboard, Annual Surface Temperature Change, World (FAO FAOSTAT, derived from NASA GISS GISTEMP).

Proposition, leaning and deceptive techniques

Proposition: "Climate change is real: global temperatures have risen meaningfully since the mid-20th century."
- Set 1 (page 1) argues the proposition is true: climate change is real.
- Set 2 (page 2) argues the proposition is false: climate change is not real, or is overstated.

Leaning toward: the first visualization from each set, the world map (page 1, top) and the year-over-year bar chart (page 2, top).

Set 1, top (world map, 2024):
- Shows only 2024, the hottest year on record, instead of a multi-year average.
- Diverging color midpoint shifted below zero (–0.3 °C), so near-zero values read as warm.
- Asymmetric, clipped color domain (–1 to +3 °C) saturates the warm end and leaves the cool arm unused.
- Area-weighted map form over-represents fast-warming high-latitude countries (Russia, Canada).
- Emotive headline with an exact, earnest callout and source to lend credibility.

Set 1, bottom (percentage of countries warmer than average, by year):
- Counts countries above their baseline instead of showing degrees, so small warming counts the same as large warming.
- A dashed 50% line stands in for "nothing is happening", so the early years read as random and the later years as a clear change.
- Red fill under the line, and only the 1965 low and the 2024 point are labelled.
- Axes, baseline and source are all accurate; the persuasion is in the metric and the framing.

Set 2, top (year-over-year change, World):
- Shows only year-over-year differences, hiding the cumulative ~1.9 °C rise.
- Omits any trend or cumulative line.
- Cherry-picked frequency stat ("28 of 63 years were cooler") that ignores accumulation.
- Symmetric y-axis to make warming and cooling look balanced.
- Accurate labels and source citation so the chart reads as credible.

Set 2, bottom (World temperature on a –5 to +5 °C axis):
- Y-axis fixed at –5 to +5 °C, so the ~1.9 °C rise fills under a quarter of the plot.
- Symmetric axis around zero, leaving a whole "cooling" half that no data point reaches.
- Calm headline and a "within 2.2 °C" callout that are true but frame the rise as small.
- Cool blue line, no trend line, no shading under the curve.
- Accurate labels, baseline, endpoints and source, so the chart reads as credible.